



GTC4Lusso USA - Cross members and diagonal braces - Latest Update: 2021-01-26

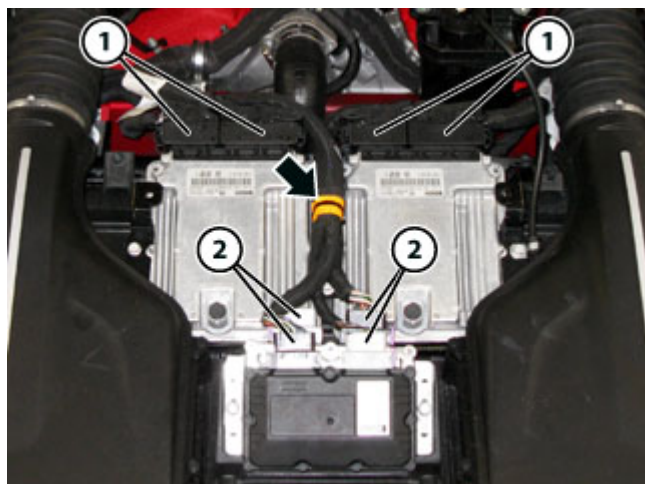
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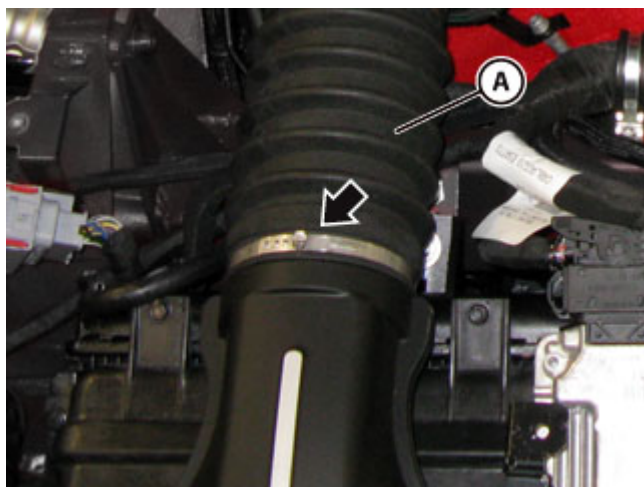
E2.04 Cross members and diagonal braces

Removing the engine compartment cross-member

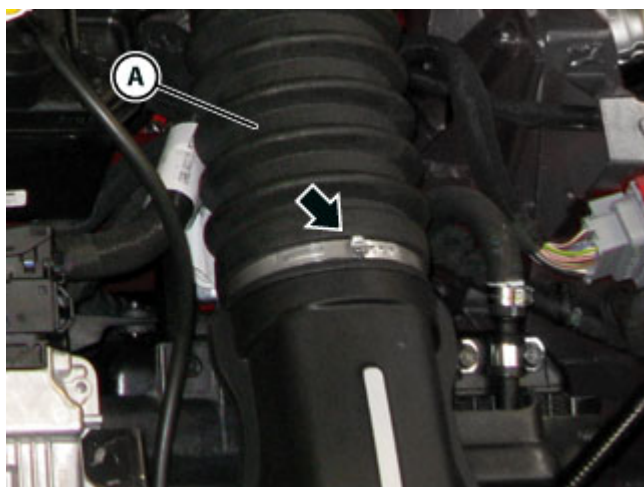
- Remove the engine compartment cosmetic shields ( E3.14).
 -  Remove the lateral cosmetic shields and the central cosmetic shield.
- Remove the air flow meters ( F3.08).



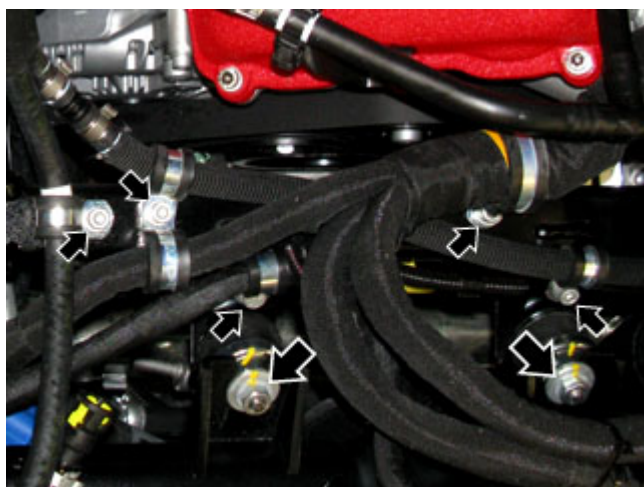
- Disconnect the connectors (1) and move the respective cables out of the way.
 -  Note which connector is connected to which ECU.
- Detach the indicated clamp.
- Disconnect the connectors (2) and move the respective cables out of the way.
 -  Note which connector is connected to which ECU.



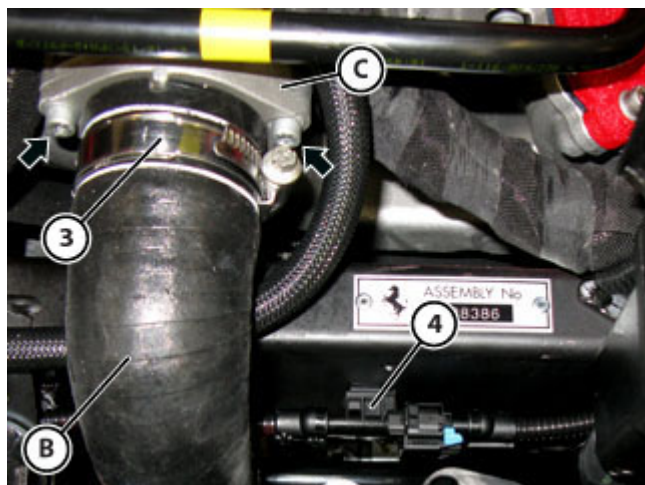
- Loosen the indicated clamp.
- Remove the bellow **(A)**.



- Loosen the indicated clamp.
- Remove the bellow **(A)**.



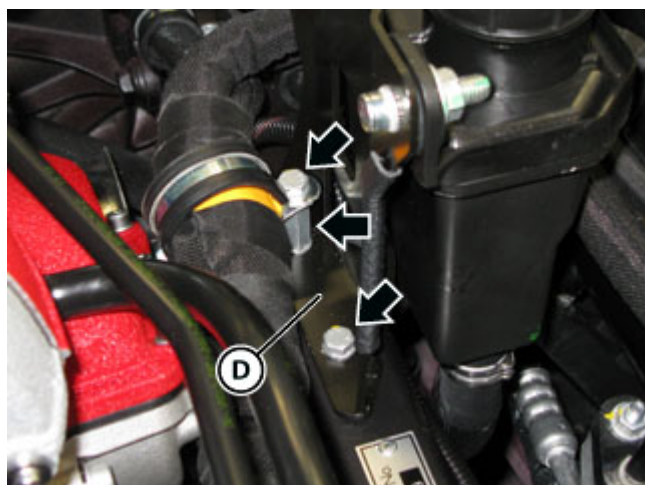
- Undo the indicated screws fastening the rubber-lined brackets.
- Undo the indicated nuts fastening the radiator assembly.



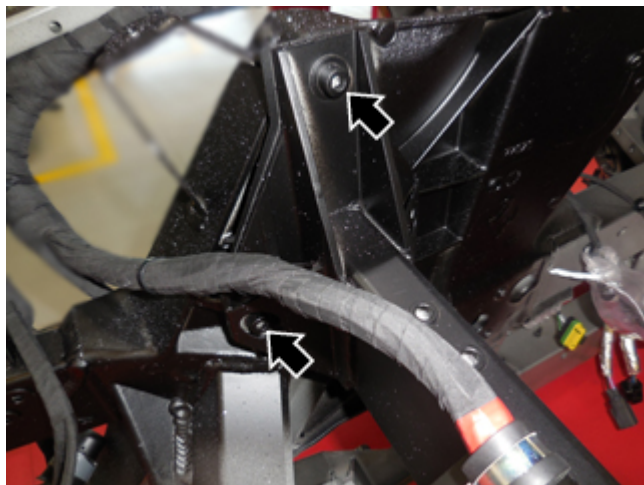
- Place a container under the hose connector **(C)** to collect any coolant contained in the pipe.
- Loosen the screw collar **(3)**.
- Disconnect the pipe **(B)**.
- Undo the screws indicated.
- Remove the hose connector **(C)**.
- Release and disconnect connector **(4)**.



- Undo the indicated screws fastening the rubber-lined brackets.
- Undo the indicated nut fastening the radiator assembly.



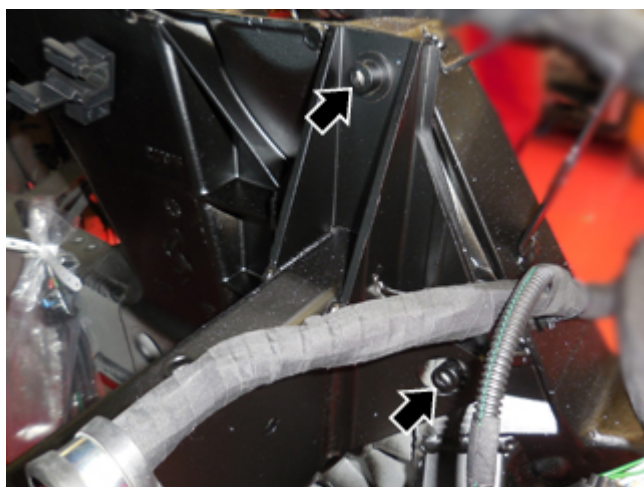
- Undo the screws indicated.
- Undo the indicated stud.
- Lift the bracket **(D)**, complete with power steering fluid reservoir.



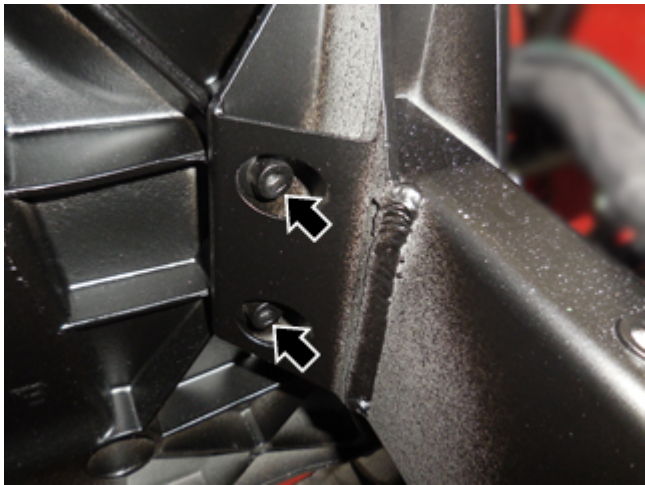
- Undo the screws indicated.



- Undo the screws indicated.



- Undo the screws indicated.



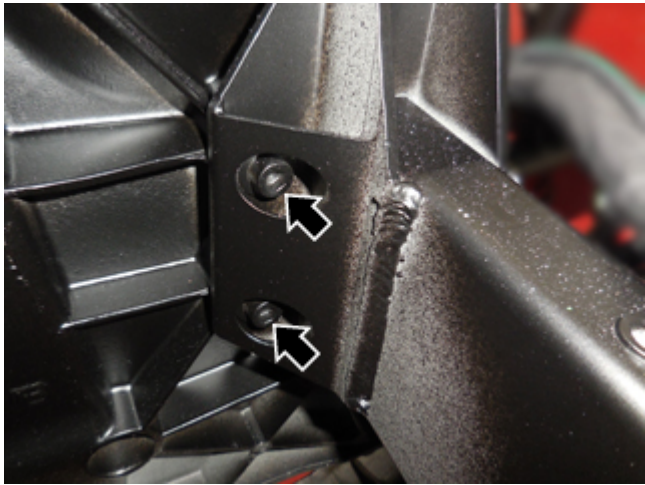
- Undo the screws indicated.

- Move the surrounding cables and pipes out of the way.
- Supporting the weight of the radiator assembly adequately, remove the cross member from the engine compartment.

Refitting the engine compartment cross-member

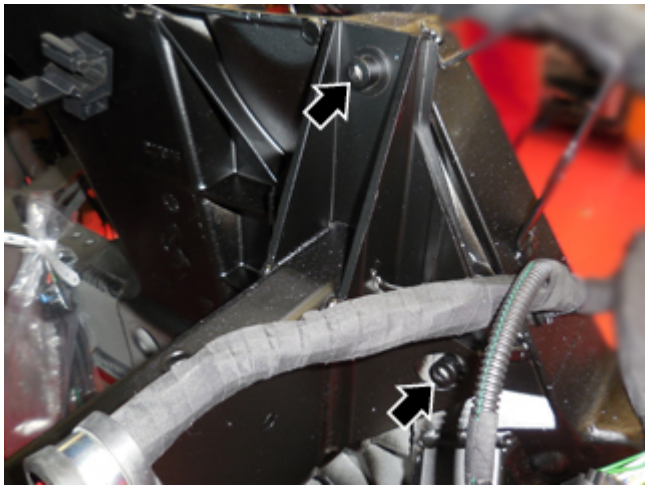
			
Tightening torque		Nm	Class
Fastening the engine compartment cross-member	Screw	30 Nm	B
	Screw	65 Nm	B
Fastening the power steering fluid reservoir bracket	Screw	9 Nm	B
Fastening the radiator assemblies	Nut	24 Nm	C
Fastening engine coolant filler pipe	Screw	9 Nm	B
Fastening engine coolant pipes	Screw collar	5 Nm	B
	Screw collar	4 Nm	B

- Supporting the weight of the radiator assembly adequately, fit the cross-member in the engine compartment and tighten the respective fastener screws.



- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	30 Nm	B



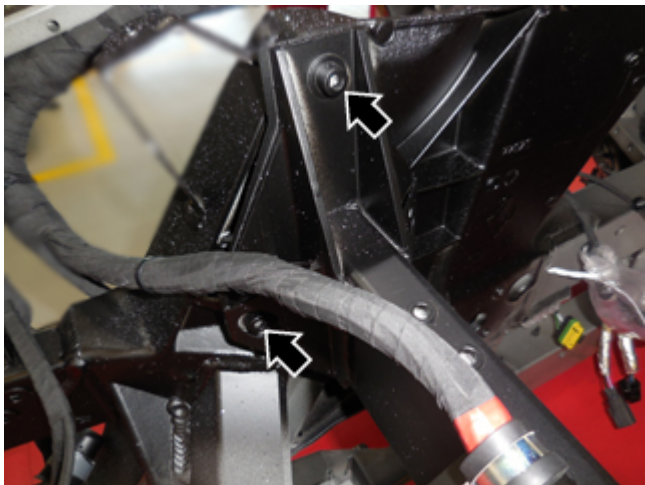
- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	30 Nm	B



- Tighten the screws indicated.

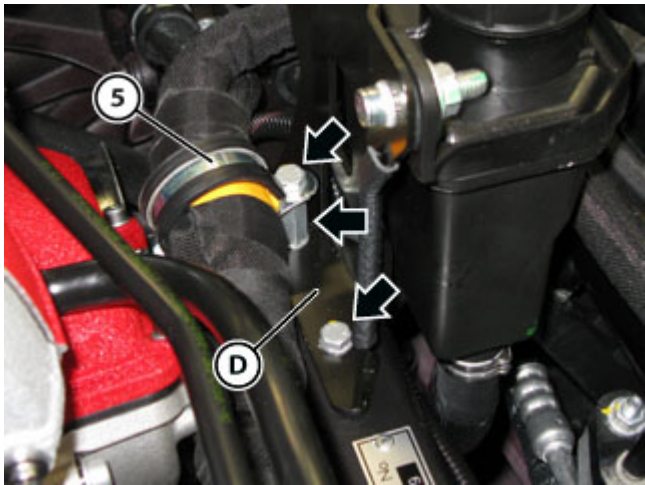
		
Tightening torque	Nm	Class
Screw	30 Nm	B



- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	30 Nm	B

- Fit the cables and pipes in their respective positions.



- Fit the bracket **(D)**, complete with power steering fluid reservoir.
- Tighten the indicated stud.
- Fit the rubber-lined bracket **(5)** and the respective cable correctly.
- Tighten the screws indicated.

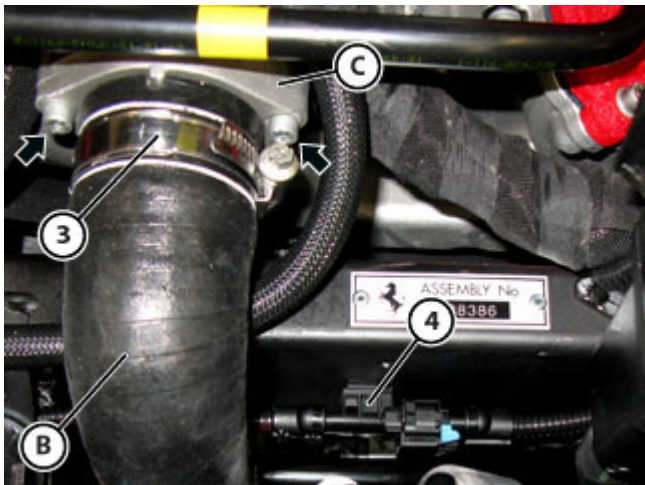
		
Tightening torque	Nm	Class
Screw	9 Nm	B



- Tighten the indicated nut fastening the radiator assembly.

		
Tightening torque	Nm	Class
Nut	24 Nm	C

- Tighten the indicated screws fastening the rubber-lined brackets.

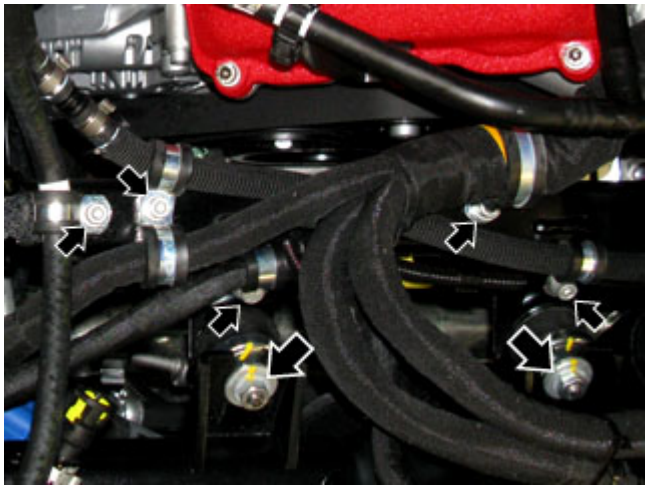


- Connect and fasten the connector **(4)**.
- Fit the hose connector **(C)**, replacing the respective seal.
- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	9 Nm	B

- Connect the pipe **(B)**, aligning the relative reference notch correctly.
- Tighten the screw collar **(3)**.

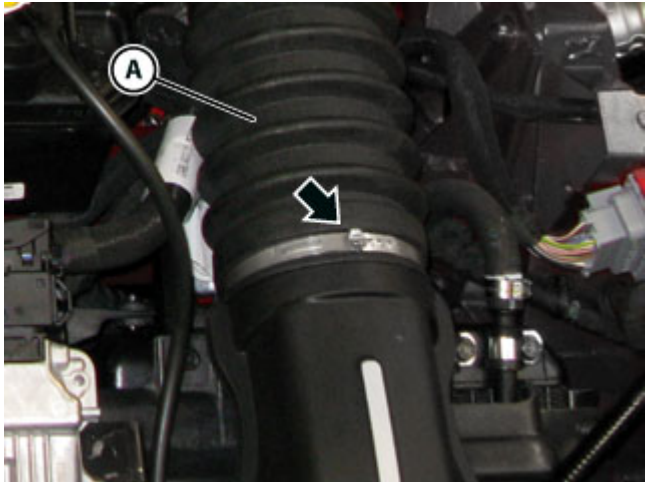
		
Tightening torque	Nm	Class
Screw collar	5 Nm	B



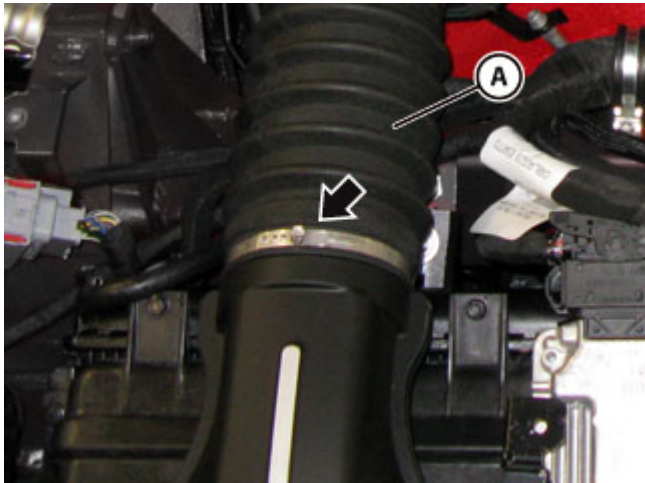
- Tighten the indicated nuts fastening the radiator assembly.

		
Tightening torque	Nm	Class
Nut	24 Nm	C

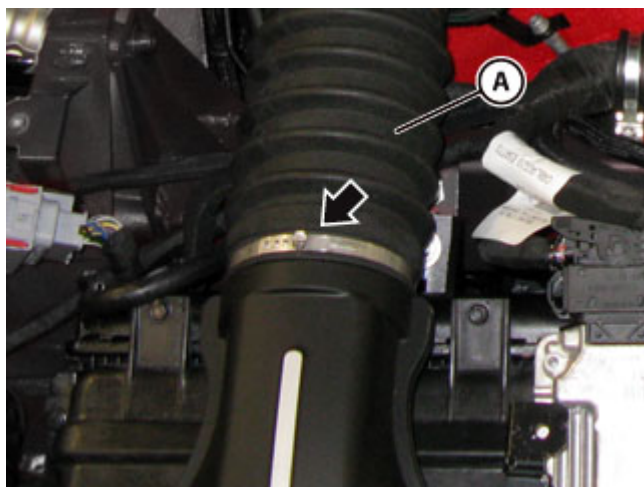
- Tighten the indicated screws fastening the rubber-lined brackets.



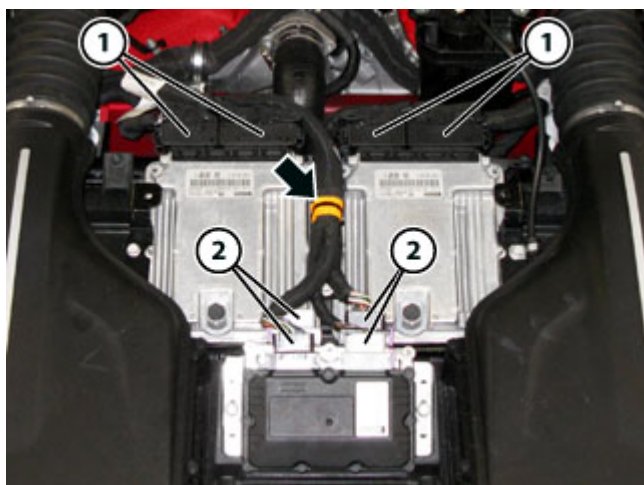
- Fit and connect the bellows (A).
- Tighten the new clamp as indicated.



- Fit and connect the bellows (A).
- Tighten the new clamp as indicated.



- Fit and connect the bellow (**A**).
- Tighten the new clamp as indicated.

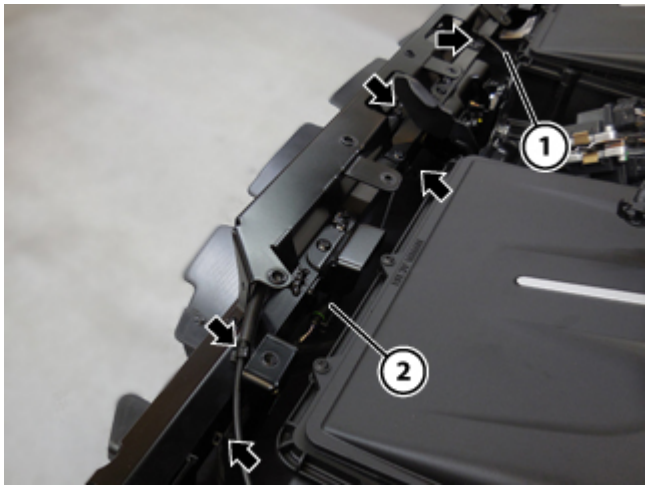


- Connect the connectors (**2**) to the respective ECUs and position the respective cables correctly.
- Connect the indicated connector.
- Connect the connectors (**1**) to the respective ECUs and position the respective cables correctly.

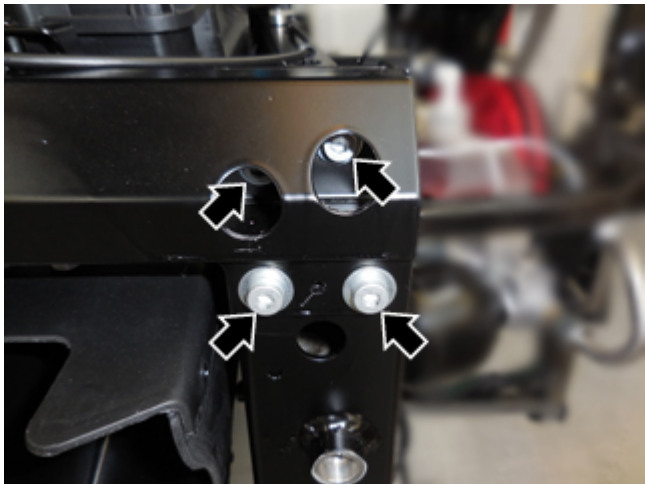
- Refit the air flow meters (🔧 F3.08).
- Check the engine cooling system fluid level (🔧 A3.07).
i This operation is necessary in the case of coolant loss during the removal procedure.
- Refit the engine compartment cosmetic shields (🔧 E3.14).
i Refit the lateral cosmetic shields and the central cosmetic shield.

Removing the upper front bumper cross member

- Disconnect the battery (🔧 F2.01).
- Remove the engine compartment cosmetic shields (🔧 E3.14).
i Remove the lateral cosmetic shields and the central cosmetic shield.
- Remove the headlights (🔧 F2.06).



- Detach the Bowden cable (1) from the engine compartment lid lock.
- Release and disconnect connector (2).
- Release the indicated clips.



- Undo the screws indicated.



- Undo the screws indicated.

- Remove the cross member.

Refitting the upper front bumper cross member

			
Tightening torque		Nm	Class
Fastening upper front bumper cross member	Screw	30 Nm	B

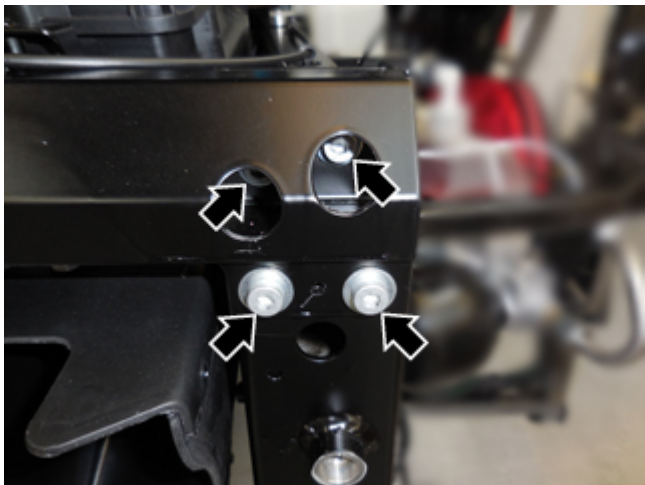
- Fit the cross member, tightening all the respective fastener screws.



- Tighten the screws indicated.



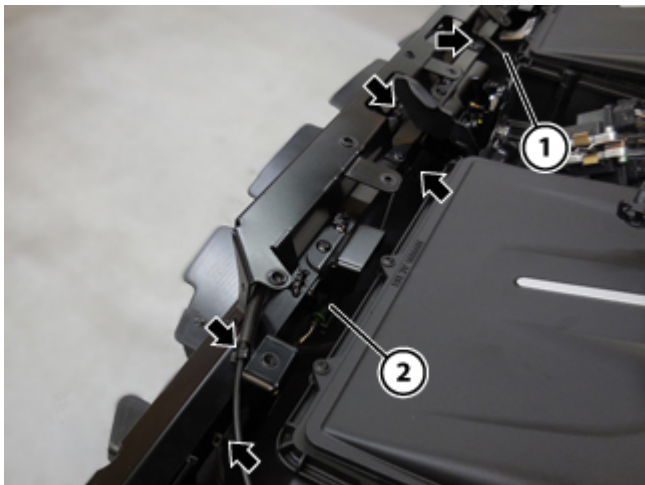
Tightening torque	Nm	Class
Screw	30 Nm	B



- Tighten the screws indicated.



Tightening torque	Nm	Class
Screw	30 Nm	B



- Attach the Bowden cable (1) to the engine compartment lid lock.
- Connect and fasten the connector (2).
- Attach the indicated clips.

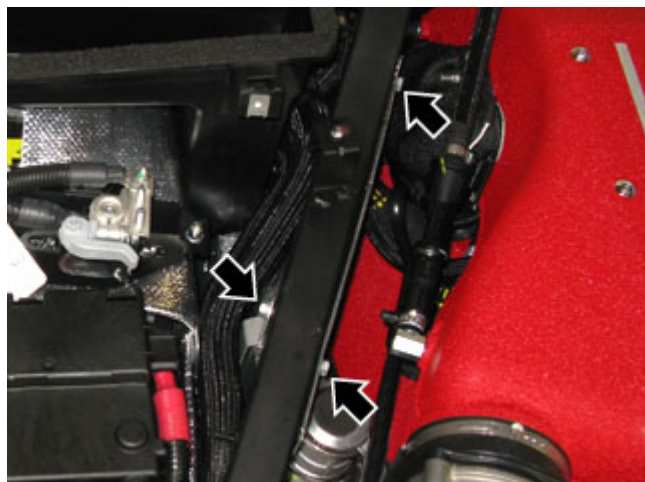
- Refit the headlight (🔧 F2.06).
- Refit the engine compartment cosmetic shields (🔧 E3.14).
- Refit the lateral cosmetic shields and the central cosmetic shield.
- Connect the battery (🔧 F2.01).

Removing the engine compartment diagonal braces

➤ Remove the engine compartment cosmetic shields (🔑 E3.14).

➤ Remove the lateral underwindscreen cosmetic shields.

Right hand diagonal brace



- Undo the screws indicated.



- Remove the rubber-lined bracket (1), undoing the respective screw indicated.
- Undo the screws indicated.

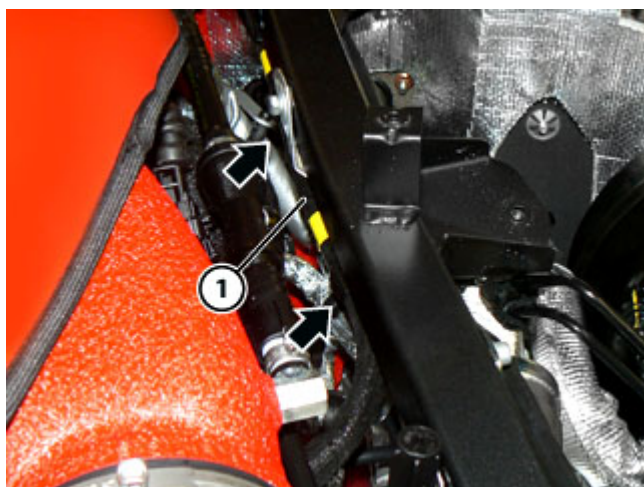


- Undo the screws indicated.

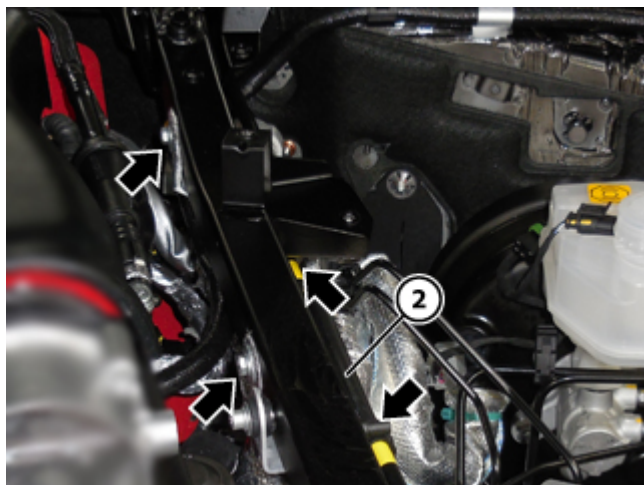
- Remove the diagonal brace.

Left hand diagonal brace

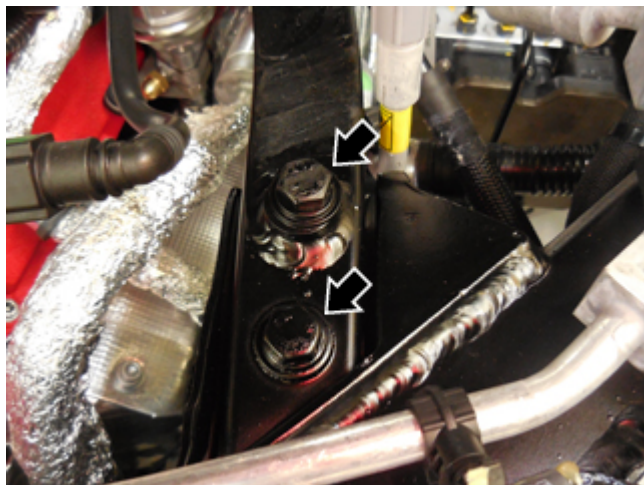
- Remove the windscreen wiper unit ( E3.10).



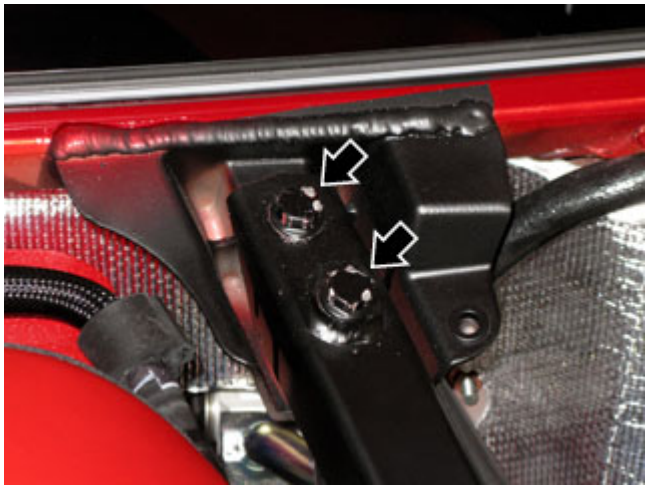
- Release the fuel pipe (1) by detaching from the indicated clips.



- Release the evaporative emissions control system pipe (2) by detaching from the indicated clips.
- Undo the screws indicated.



- Undo the screws indicated.



- Undo the screws indicated.

- Remove the diagonal brace.

Refitting the engine compartment diagonal braces

			
Tightening torque		Nm	Class
Fastening the engine compartment diagonal braces	Screw	34 Nm	B
Fastening engine compartment heat shields	Screw	9 Nm	B
Fastening the TKG solenoid valve pipe mounting bracket	Screw	9 Nm	B
Fastening Freon pipes	Nut	16 Nm	B
	Screw	9 Nm	B
	Nut	24 Nm	B
	Screw	5 Nm	B

Right hand diagonal brace

- Fit the diagonal brace.



- Tighten the screws indicated.



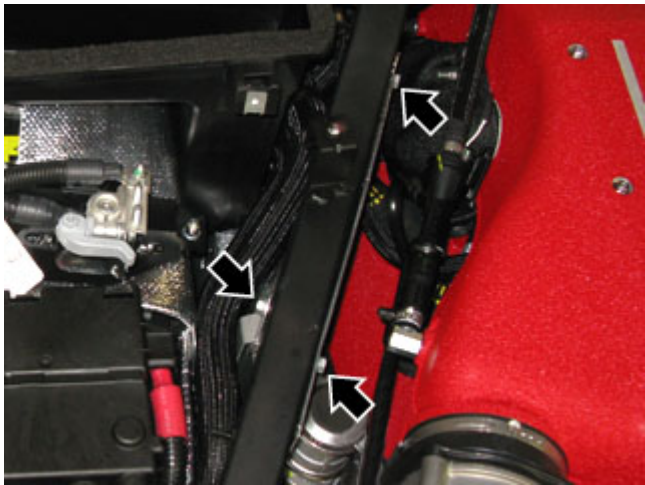
- Fit the rubber-lined bracket (1), tightening the respective screw indicated.
- Tighten the screws indicated.

- Definitively tighten the inner screws fastening the diagonal brace.

		
Tightening torque	Nm	Class
Screw	34 Nm	B

- Definitively tighten the outer screws fastening the diagonal brace.

		
Tightening torque	Nm	Class
Screw	34 Nm	B

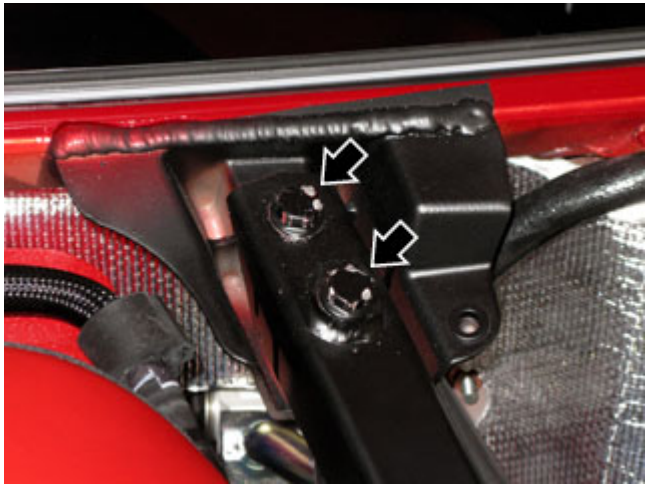


- Tighten the screws indicated.

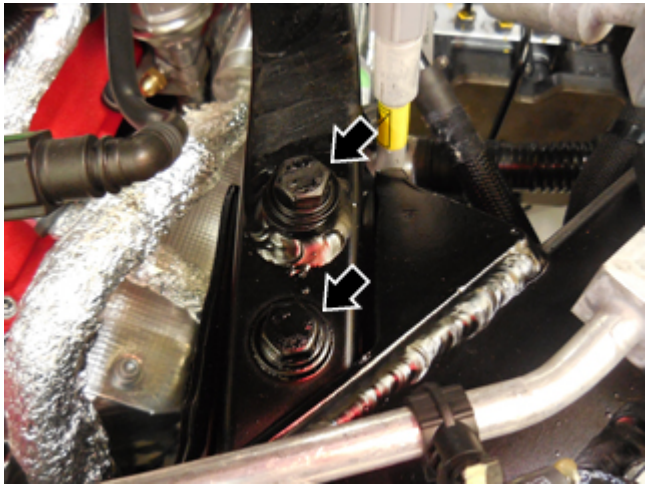
		
Tightening torque	Nm	Class
Screw	9 Nm	B
Screw	9 Nm	B

Left hand diagonal brace

- Fit the diagonal brace.



- Tighten the screws indicated.



- Tighten the screws indicated.

- Definitively tighten the inner screws fastening the diagonal brace.

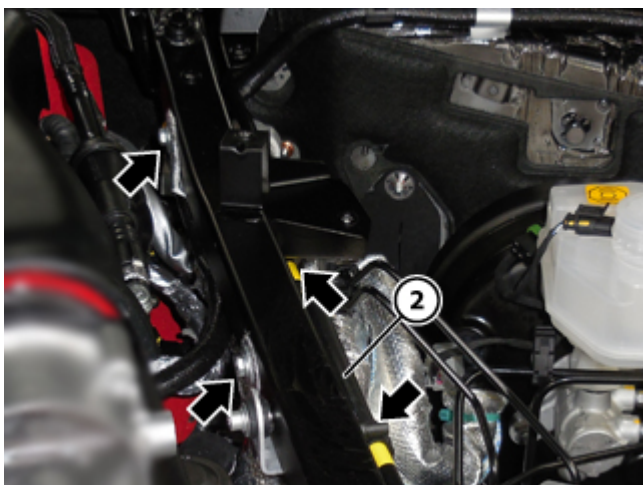


Tightening torque	Nm	Class
Screw	34 Nm	B

- Definitively tighten the outer screws fastening the diagonal brace.



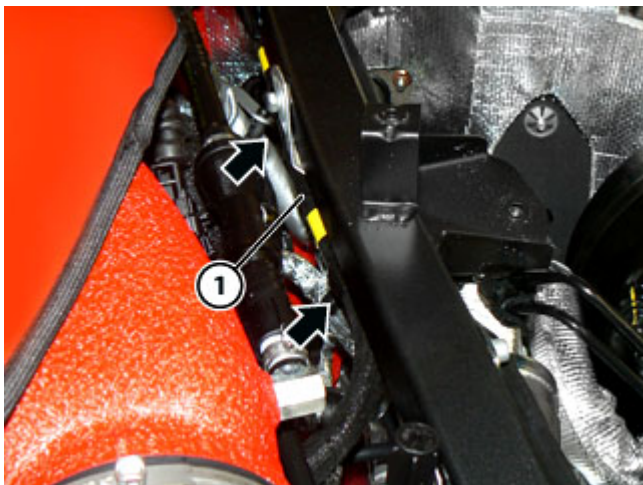
Tightening torque	Nm	Class
Screw	34 Nm	B



- Secure the evaporative emissions control system pipe (2) by fastening the indicated clips.
- Tighten the screws indicated.



Tightening torque	Nm	Class
Screw	9 Nm	B
Screw	9 Nm	B



- Secure the fuel pipe (1) by fastening the indicated clips.

- Refit the windscreen wiper unit ( E3.10).

- Refit the engine compartment cosmetic shields ( E3.14).

